Glow Swarm - Technical Sheet

Tech used

One self-contained HTML file (about 30 KB), no libraries, no external assets. HTML5 Canvas 2D with additive blending and shadow glow for neon rendering, capped at 2x device pixel ratio. Web Audio API for all sound effects and the synth music, generated live. localStorage for saves. Pointer, touch and keyboard input.

Architecture

A pure game-logic core (newGame, step, choose) with no DOM access, shared by the browser game and a headless Node.js simulator. The game shell handles input, rendering, audio, menus and saves. Variable frame time clamped to 50 ms with sub-stepping for slow frames.

Systems

5 weapons and 6 passives with levels, 5 evolutions, weighted 3-card level-up draws (evolutions always offered first). 3 enemy types plus 2 bosses; HP, speed, damage and spawn rate scale with time; minute-mark surges. Gem magnet and gem merging (cap of 300 gems), enemy cap of 260, distant enemies recycle near the player. Invulnerability frames after hits. Meta progression with 4 permanent upgrades. Music tempo rises over the run and the filter opens during boss fights.

Graphics and sound

Parallax starfield and neon grid, glowing animated enemy shapes, hit flashes, particle bursts, floating damage numbers, screenshake, screen flash, low-HP red pulse, boss arrow and health bar, event banners. Synth bass line, kick drum and lead melody plus 14 sound effects, rate-limited so big fights do not get too loud.

Audit 1: Balance (simulated play)

A bot player (it flees nearby enemies, circles, and picks upgrades by priority) played 30 seeded runs in Node.js on the real game core.

Before: average survival 4:28, median 4:13, 0 of 30 runs won, 7 of 30 reached the first boss, average level 6.6. The XP curve was far too steep and the game was unwinnable.

Fixes: flatter XP curve (growth exponent 1.12), gem magnet 95 px, gentler enemy HP growth (0.08), contact damage 8.

After: average survival 6:58, median 5:29, 11 of 30 wins (37%), 20 of 30 reached the first boss, average level 21.3, about 1,575 kills per run. A tested middle option (19 of 24 wins) was rejected as too easy.

Audit 2: Bugs

Syntax-checked the full combined script with Node. Reviewed the state flow (menu, play, level up, pause, end, shop). Fixes: level-up cards ignore taps for the first 350 ms so a thumb still dragging cannot pick a card by accident; the joystick is released when a level-up opens; the game auto-pauses when the tab or app is hidden; audio is optional, so browsers without Web Audio still run; the music

clock resets after long pauses so it never plays a burst of catch-up notes; gem and enemy counts are capped to keep phones smooth.

Audit 3: Player experience

Added a drag-to-move hint that hides on first touch, a floating joystick that follows the thumb, evolution cards highlighted in gold, NEW / LV tags on cards, a pause screen listing the loadout, a boss health bar and off-screen boss arrow, a red edge pulse under 30% HP, and an end screen with time, level, kills, shards earned and a new-best badge.

Limits

Tested with simulated play and code review, not on a physical phone.